

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) accelerates generative training by distilling representations from a pretrained vision encoder into intermediate diffusion features. We make three contributions that clarify why this recipe works and how to improve it. First, we show that target encoders are not best selected by *global* semantic strength or semantic performance alone: across 27 vision encoders and multiple diffusion model scales, ImageNet-1K accuracy is a weak predictor of generation quality under REPA. Second, we introduce a simple Spatial Structure Metric (SSM) family based on pairwise cosine similarity between patch tokens, and show that these spatial-structure scores explain generation FID substantially better than linear probing accuracy. Third, we translate the analysis into iREPA, a minimal spatially focused modification to REPA that replaces the standard MLP projection with a lightweight convolutional projection and applies spatial normalization to the external representation. Although iREPA requires <4 lines of code, it consistently improves REPA convergence across vision encoders, model sizes, and training variants including REPA, REPA-E, MeanFlow, and JiT. Together, these contributions reframe representation alignment for generation as a spatial-transfer problem and provide both a diagnostic signal and a simple intervention for selecting and using target representations.

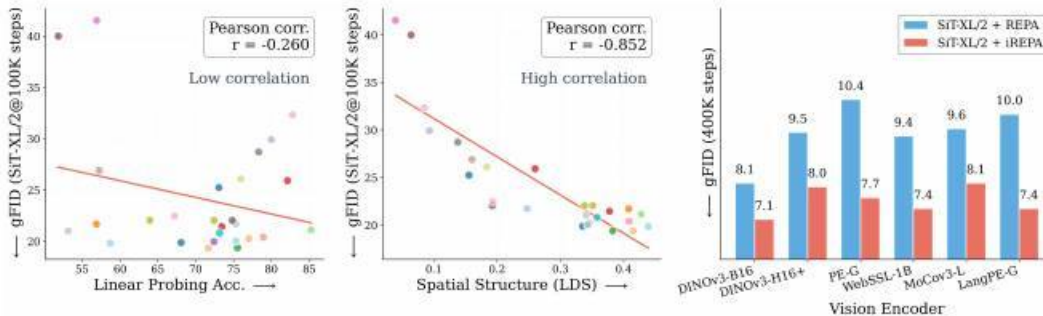


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis across 27 diverse vision encoders. Surprisingly, contrary to the prevailing wisdom, we find that spatial structure, rather than global performance (measured by linear probing accuracy), drives the generation performance of a target representation. *Right:* We further study this by introducing two simple modifications to accentuate the transfer of spatial features from target representation to diffusion model. Our simple approach consistently improves the convergence speed of REPA across diverse settings.

1 INTRODUCTION

Representation alignment is now a practical recipe for accelerating diffusion transformer training (Ma et al., 2024; Peebles & Xie, 2023): distilling pretrained visual features into intermediate diffusion features improves convergence and final generation quality (Yu et al., 2024; Leng et al., 2025a). The open question, and the central contribution of this paper, is not whether alignment helps but *what* useful signal is being transferred. We frame the manuscript around a concrete reviewer-facing alternative: REPA may work because the target encoder carries stronger global semantic information,



Figure 2: **Motivating examples — spatial structure matters.** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. *Left:* PE-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to SpatialPE-B (Bolya et al., 2025). *Right:* Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Spatial Self-Similarity:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

often measured by linear probing, or because it preserves spatial structure in the relationships among patch tokens.

This distinction matters because it changes the selection rule for target encoders in generative training. A common assumption is that a better alignment target should also be a stronger ImageNet-1K representation, with validation accuracy used as a proxy for global semantic quality (Oquab et al., 2024; Chen et al., 2021) (e.g., stronger image classification). Under that view, target representations with better ImageNet performance should lead to better generation (Yu et al., 2024), and improved linear-probe accuracy of diffusion features is treated as evidence that representation alignment transfers global semantics (e.g., class-level information). The following quote from REPA (Yu et al., 2024) makes the assumption explicit and motivates our first contribution:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

We challenge this interpretation with a systematic study of what makes a target representation useful for generation. The study spans diverse vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024). The paper’s contribution structure follows directly from the evidence: global accuracy is often a poor selection criterion, spatial structure provides a stronger explanatory signal, and preserving spatial transfer leads to a simple improvement over standard REPA.

Contribution 1: Higher validation accuracy does not imply a better representation for generation. Contrary to the standard assumption, stronger global semantic performance, measured by ImageNet-1K linear probing accuracy, does not reliably identify better REPA targets. PE-Spatial-B, a spatially tuned model derived from PE-Core-G (Bolya et al., 2025), has much lower patch-token validation accuracy than PE-Core-G (53.1% vs. 82.8%)¹, yet produces better generation with REPA (Figure 2). WebSSL-1B (Fan et al., 2025) gives a second counterexample: it has much better global performance than PE-Spatial-B (76.0% vs. 53.1%) but worse generation. Section §2 turns these examples into the paper’s first contribution by showing that global accuracy repeatedly fails as a predictor for representation alignment.

Contribution 2: Spatial structure, not global information, determines generation performance. To make the alternative measurable, we introduce and evaluate simple metrics for spatial self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3). We correlate REPA generation FID with both linear probing accuracy and spatial structure across 27 vision encoders and 3 diffusion model sizes (SiT-B, SiT-L, SiT-XL). The resulting diagnostic is much clearer for reviewers and practitioners: spatial structure metrics correlate far more strongly with generation FID (Pearson $|r| > 0.852$) than ImageNet-1K validation accuracy ($|r| = 0.26$). Controlled evidence supports the

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

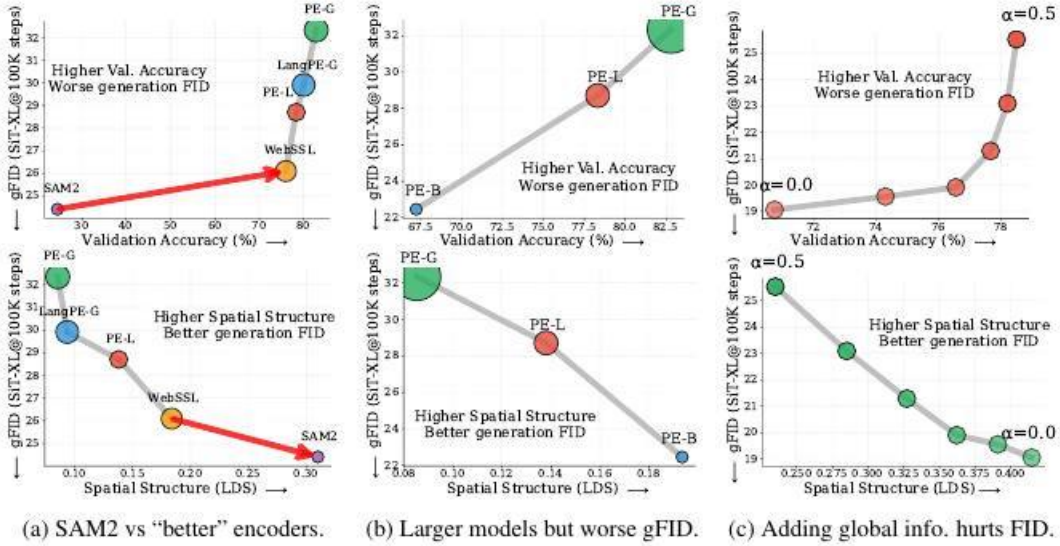


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Several trends showing global performance does not correlate well with generation FID when using REPA. (a) SAM2-S with only validation accuracy of 24.7% results in better generation performance with REPA compared with other models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within same family can have better validation but worse generation performance. (c) Adding global information to patch tokens via CLS token improves global performance but hurts generation. **Bottom row:** We show that spatial structure rather than global performance provides a better indicator for generation. Please see §3 for large-scale detailed analysis across different spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

same conclusion: SAM2 can outperform encoders with $\sim 60\%$ higher ImageNet accuracy (Fig. 3), and even classical spatial features such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005) provide useful REPA gains (§3).

Contribution 3: Accentuating spatial transfer improves convergence. The empirical findings suggest a concrete intervention: if spatial structure is the useful signal, REPA should preserve and emphasize that signal during alignment. We therefore introduce two minimal modifications (<4 lines of code): (1) a spatial regularization layer that boosts spatial contrast in the target representation, and (2) a convolutional projection that replaces the standard MLP used to map diffusion features to target dimensions, because the MLP loses spatial information (Figure 6a). The resulting method, iREPA, consistently improves REPA convergence across target encoders, model sizes, and training variants such as REPA-E (Leng et al., 2025a) and MeanFlow (Geng et al., 2025) w/ REPA.

We summarize the paper’s contributions as follows:

- We provide a large-scale empirical analysis showing that spatial structure, rather than global semantic accuracy, drives the effectiveness of representation alignment for generation.
- We introduce a straightforward and fast-to-compute Spatial Structure Metric (SSM) family that predicts downstream FID substantially better than linear probing scores.
- We propose iREPA, a minimal spatially focused REPA modification that improves convergence across target representations, diffusion model sizes, and training variants including REPA, REPA-E, MeanFlow w/ REPA, and JiT w/ REPA.

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

This section establishes the paper’s first empirical contribution: global semantic performance, measured by ImageNet-1K accuracy (*e.g.*, linear probing), is not a reliable indicator of how useful a target representation will be for REPA. We present the evidence as a sequence of stress tests for the global-accuracy hypothesis. Each test is hard to reconcile with global accuracy alone, and each motivates the spatial-structure explanation developed in §3.

Recent vision encoders. We begin with current vision encoders that a practitioner might reasonably choose as REPA targets, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al.,

2025), and DINOv3 (Siméoni et al., 2025). The clearest counterexample is PE-Spatial-B (80M), a small spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). Figure 2 shows the contribution-relevant contrast: PE-Core-G has much higher ImageNet-1K accuracy (82.8% vs 53.1%), yet is worse as a REPA target (FID 32.3 vs 21.0). WebSSL-1B (1.2B) repeats the pattern, reaching much higher ImageNet-1K accuracy than PE-Spatial-B (76.0% vs 53.1%) but worse REPA generation (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. A stronger stress test asks whether REPA can benefit from an encoder with very limited global classification information. We therefore evaluate the SAM2-S vision encoder (46M) (Ravi et al., 2024), which has only 24.1% validation accuracy (Fig. 3a). Despite this low global score, SAM2-S achieves better FID under REPA than encoders with substantially higher ImageNet-1K accuracy, including PE-Core-G (82.8%). This result sharpens the first contribution: strong global recognition quality is not necessary for strong alignment gains.

Larger models within the same encoder family can have similar or worse generation performance. If global accuracy were the primary driver, scaling an encoder family should generally make it a better REPA target. Instead, larger variants often produce similar performance (DINOv2) or even worse generation (PE, Cradio), as shown in Fig. 3b. Notably, (Yu et al., 2024) report a similar DINOv2 pattern and hypothesize that all DINOv2 models share similar representations because they are distilled from DINOv2-g. In §3, we show that spatial structure provides a more direct explanation of this scaling trend.

Adding more global information can hurt generation. To isolate global information within a fixed encoder, we inject CLS-token semantics into local patch tokens using DINOv2. The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. Figure 3c gives the most controlled evidence for the contribution: as α increases from 0 to 0.5, linear probing accuracy improves monotonically from 70.7% to 78.5%, but generation quality deteriorates, with FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$.

Together, these observations establish the first finding and set up the next contribution. Global performance alone does not explain REPA quality; §3 therefore introduces spatial structure as a measurable alternative signal for target-representation quality.

- **Finding 1.** *Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.*

3 SPATIAL STRUCTURE MATTERS MORE

This section presents the second contribution: a simple spatial-structure view that predicts when a target representation will work well for REPA. Building on the failures of global accuracy in §2, we test the hypothesis that generation performance depends on spatial structure in the target representation. We operationalize the hypothesis with spatial self-similarity metrics (Shechtman & Irani, 2007), show that these metrics correlate much more strongly with gFID than ImageNet-1K accuracy, and use them to explain the counterexamples that motivated the paper.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define the two quantities needed to make the spatial contribution measurable:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extracted patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

The metric contribution is to score how self-similarity $K_{\mathcal{X}}$ changes with lattice distance d between patch tokens. Larger values indicate stronger spatial organization: nearby patches remain more similar to one another than distant patches. Our default score is a simple correlogram contrast (local

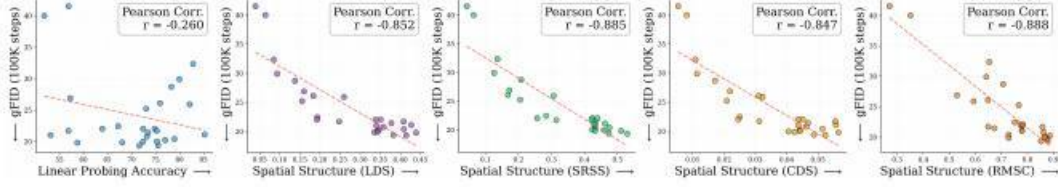


Figure 4: SSM contribution: spatial structure predicts generation quality better than linear probing. Across 27 vision encoders with SiT-XL/2 and REPA, linear probing has weak correlation with FID (Pearson $|r| = 0.260$), while spatial metrics provide much stronger reviewer-facing predictors: LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$). See Fig. 10 for detailed plots with encoder labels.

vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t,t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t,t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ here, and the correlation is robust to the exact threshold choices. Appendix B gives the exact details and reports alternative spatial metrics (CDS, SRSS, RMSC), which also provide effective explanatory signals (see Fig. 4).

Spatial structure correlates much higher with generation performance than linear probing. We next test the metric at scale across 27 diverse vision encoders. Figure 4 provides the main evidence for the second contribution: the commonly used linear probing score has very weak correlation with generation performance (Pearson $|r| = 0.26$), whereas all SSM metrics show much stronger correlation (Pearson $|r| > 0.85$).

Generalization across model scales. We also verify that the spatial signal is not tied to one diffusion model size. Across SiT-B, SiT-L, and SiT-XL in Figure 5, linear probing remains weakly correlated with generation quality ($|r| < 0.306$), while spatial structure remains highly correlated ($|r| > 0.826$). This scale check makes SSM a general diagnostic rather than a single-model observation.

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure can explain previously unexplained trends. The value of SSM is not only predictive correlation; it also makes the counterexamples from §2 interpretable. Global performance (ImageNet-1K accuracy) fails as a selection rule for representation alignment, while spatial metrics consistently identify the better REPA targets.

(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%) but performs worse as a REPA target (FID 32.3 vs 22.0). The same comparison in Figure 2 also gives the spatial explanation: despite lower global performance, PE-Spatial-B has much stronger spatial structure than PE-Core-G, matching its better generation performance.

(2) *SAM2 outperforms “better” vision encoders:* §2 shows that SAM2, despite much lower ImageNet-1K accuracy (24.1%; e.g., weak classification performance), yields better generation than encoders with $\sim 60\%$ higher accuracy. Fig. 3a makes this result consistent with the contribution: SAM2 has better spatial structure.

(3) *Larger models in the same encoder family underperform:* Fig. 3b shows that larger models within an encoder family can improve ImageNet-1K accuracy while degrading spatial structure, leading to worse generation under REPA. This spatial explanation also aligns with observations from Yu et al. (2024), where larger DINOv2 models can have similar or worse generation performance.

(4) *Adding global information to patch tokens via CLS token hurts generation:* Figure 3c shows that mixing the CLS token into patch tokens increases global performance but reduces spatial contrast among patch tokens. Tokens from unrelated regions, such as foreground objects and background, become overly similar; the resulting loss of spatial structure explains the worse generation performance.

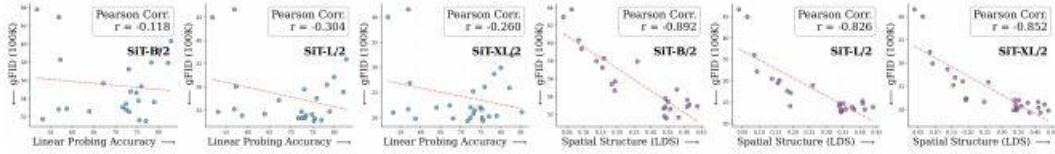
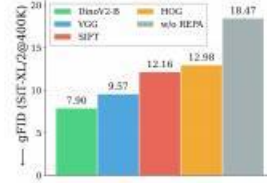
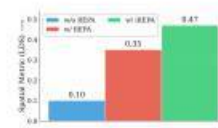


Figure 5: **SSM contribution across model scales.** Spatial structure (right) consistently correlates more strongly with gFID than linear probing (left), showing that the SSM diagnostic is not specific to one diffusion model size.

If spatial structure matters more, can SIFT or HOG features work for REPA? Surprisingly, yes. Although classical spatial features are not the best targets, SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) all improve REPA. This sanity check sharpens the contribution: representation alignment can benefit from spatial features even without strong additional global semantics.



Can spatial metrics explain gains from REPA? Yes. The introduced spatial metrics explain both the gains from standard REPA and the improved training recipe, iREPA, introduced next in §4. This closes the diagnostic contribution and motivates the method contribution.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

This section turns the spatial-structure findings into the paper’s third contribution: a minimal REPA modification that more directly transfers spatial information from the teacher vision encoder to the student diffusion transformer. The method, iREPA, keeps the original REPA objective but changes two implementation details that otherwise weaken spatial structure during alignment.

Convolutional projection layer instead of MLP. Standard REPA uses a 3-layer MLP projection to map diffusion features into the dimensionality of the external representation. Figure 6a shows the issue for our spatial-transfer contribution: the projection is lossy and reduces spatial contrast between patch tokens. We therefore replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) that operates directly on the spatial grid. The convolutional projection keeps the intervention simple while preserving local spatial relationships through its inductive bias.

Spatial normalization layer. Similar to Siméoni et al. (2025), we find that patch tokens of pretrained vision encoders contain a substantial global component, as indicated by high linear probing scores for the mean of patch tokens (Figure 14). Figure 6b further shows why this matters for iREPA: mixing this global information (the mean of patch tokens) into patch tokens can improve global performance but reduces spatial contrast among individual tokens. As a result, tokens from different regions, such as a foreground object and background, can become highly similar.

Given the evidence in §3, iREPA intentionally trades some global patch-token information for stronger spatial contrast. The goal is to provide REPA with a cleaner spatial signal, measured by pairwise similarity between patch tokens. We implement the intervention with a simple spatial normalization layer (Ulyanov et al., 2016) applied to target-representation patch tokens:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ for numerical stability.

4.1 EXPERIMENTS

The experiments evaluate whether the spatial contribution is practically useful, not merely explanatory. On Imagenet 256×256 , we ask three questions that map directly to the claimed value of iREPA:

- Does iREPA improve diffusion transformer convergence over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)

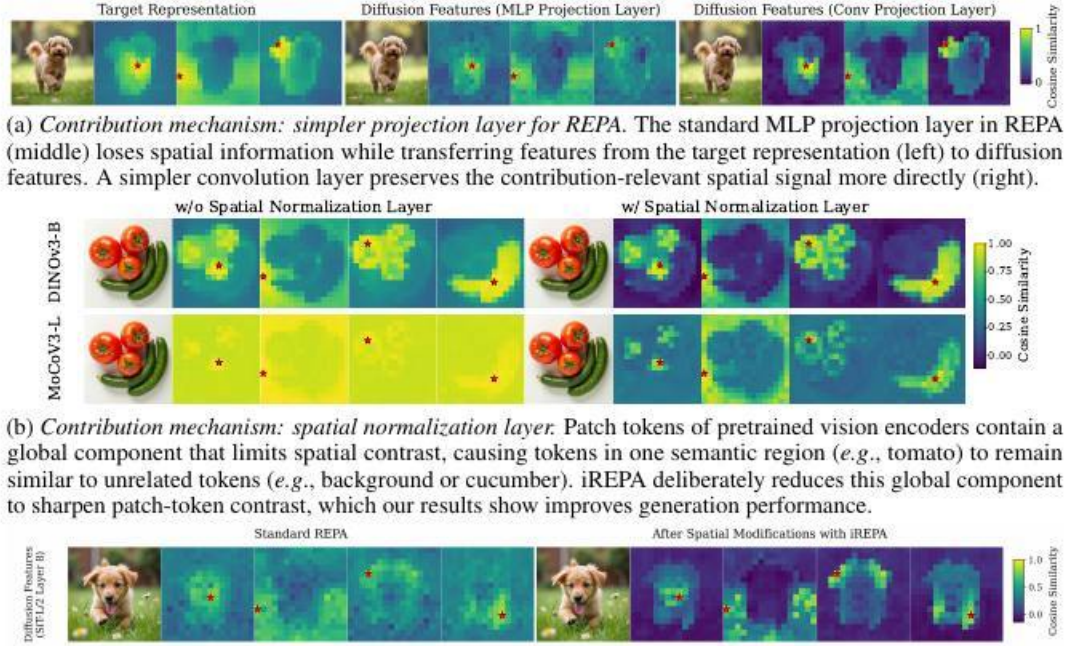
Algorithm 1 Summary of key iREPA modifications to standard REPA training

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# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)

```



(c) Overall contribution of iREPA. The improved training recipe increases the spatial structure of diffusion features.

Figure 6: iREPA design motivation. Two minimal modifications make the spatial-transfer contribution concrete: a convolutional projection preserves local structure, and spatial normalization increases target-feature contrast. All results are reported with SiTL/2 with REPA at 100K.

- Does iREPA scale across model sizes and remain effective under different training settings? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA transfer to recent representation-alignment variants such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), as well as pixel-space diffusion with JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We first test the main practical claim: spatially focused alignment should converge faster than baseline REPA. Across diverse vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2), Fig. 7 shows that iREPA consistently reaches better FID sooner than REPA. This result connects the explanatory contribution to a direct training benefit.

Target representation. We next ask whether the gain depends on a specific teacher encoder. Fig. 8 and Table 4 show that iREPA improves generation quality across the evaluated vision encoders. This supports the method contribution as a broadly useful spatial modification rather than an encoder-specific tuning. Additional comparisons across all 27 encoders are provided in Appendix C, E.

Encoder size. Table 1a tests whether iREPA remains useful as the target encoder size changes from PE-B (90M) to PE-L (320M) and PE-G (1.88B). iREPA improves performance for all three encoder sizes, and the percentage improvement increases with encoder size (22.2% for PE-B, 38.8% for PE-L, 39.6% for PE-G). This supports the contribution that spatial transfer remains important even when larger encoders provide more global capacity.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

(a) Vision Encoder size

Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

(b) Model size

Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** We find that iREPA leads to consistent gains over baseline REPA across diverse training settings. Unless otherwise specified all results are reported using Dinov2 as encoder, SiT-XL/2, 100K steps and vanilla-REPA as baseline.

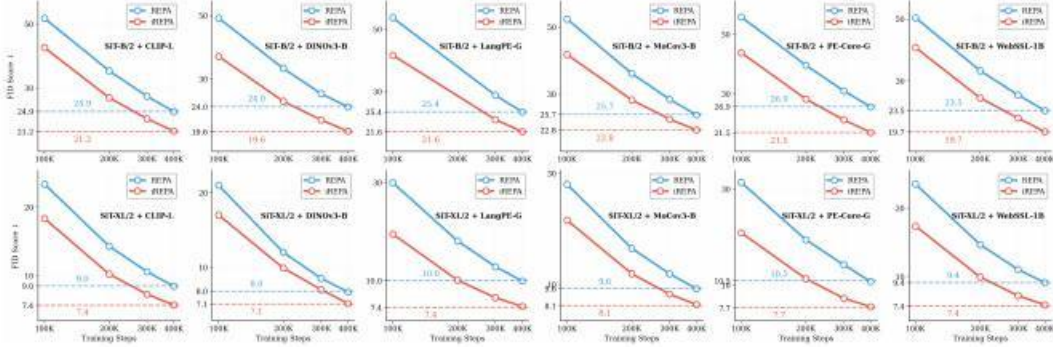


Figure 7: Accelerating spatial features helps consistently improve convergence speed.

Scalability. Table 1b evaluates iREPA across diffusion model scales. The spatial improvements consistently improve performance, with larger percentage gains for larger models. Thus, the method contribution scales with model size rather than being confined to a small-model regime.

Encoder depth. Table 1c evaluates whether the contribution depends on a particular alignment layer. Using SiT-B/2 at 100K iterations with DINOv3-B as the target representation, iREPA improves over baseline REPA across different alignment depths.

Ablation on different components. Table 2 isolates the two iREPA components. Both spatial normalization and the convolutional projection layer improve generation quality over baseline REPA, and the best result is obtained when both are used together. This ablation makes the contribution structure explicit: each component helps preserve or emphasize spatial structure.

Training recipe. We then test whether iREPA is tied to the original REPA recipe. Table 3 applies the spatial modifications on top of REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), showing convergence gains across these training recipes. This result broadens the contribution from a single implementation tweak to a reusable spatial-alignment principle.

Classifier-free guidance. Table 4 evaluates iREPA with and without CFG across different vision encoders at 400K training iterations. Across encoders, iREPA improves convergence in both settings, indicating that the spatial gains persist under guided sampling.

Pixel-space diffusion (JiT). Finally, we test whether the contribution extends beyond latent diffusion transformers. Figure 9 compares REPA and iREPA on pixel-space diffusion with JiT-B (Li & He, 2025b), following the JiT setup (Li & He, 2025b). Accentuating spatial information consistently improves convergence across vision encoders, including DiNov2, DiNov3, PE, etc..

- **Finding 3.** Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).

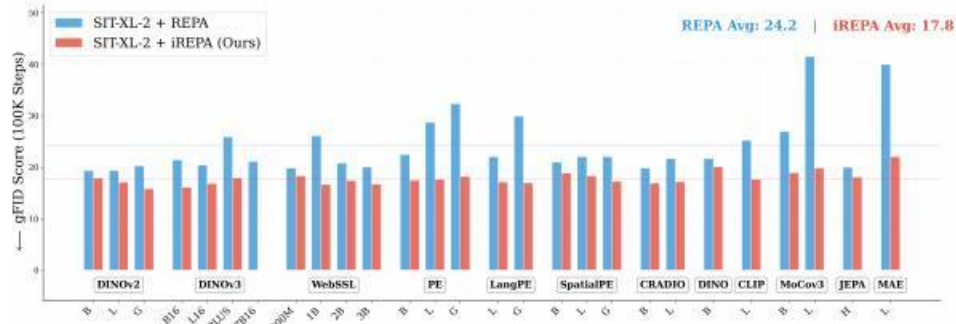


Figure 8: **Variation in target representation.** Across all 27 vision encoders, we find that accentuating transfer of spatial features from target representation to diffusion features (iREPA) leads to consistent improvements in convergence speed. See Appendix C for more results across diverse settings.

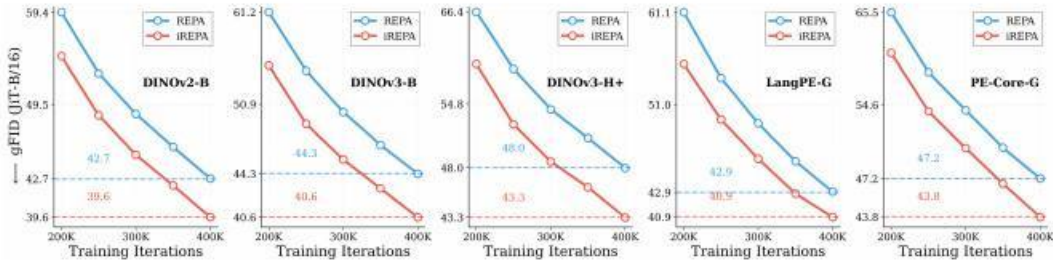


Figure 9: **Pixel-space diffusion contribution.** FID convergence curves compare REPA and iREPA with pixel-space JiT (Li & He, 2025b) across vision encoders. The consistent speedups show that the spatial-transfer contribution is useful beyond latent diffusion transformers. All results are reported with JiTB/16 (Li & He, 2025b), 256 batch size and without classifier-free guidance. Refer Table 10 for further results.

5 RELATED WORK

We position the three contributions relative to the most relevant prior work here and provide a more detailed discussion in Appendix H.

Representation alignment for generation. Recent work increasingly uses external representations to improve diffusion model training (Pernias et al., 2023; Fuest et al., 2024). In particular, (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) show that aligning internal diffusion features with clean-image features from a pretrained vision encoder can substantially improve performance. Related extensions apply this idea to video generation and 3D generation (Zhang et al., 2025; Wu et al., 2025), and to unified models (Ma et al., 2025a). Our contribution is complementary: rather than proposing only another alignment variant, we identify which aspect of the target representation makes alignment effective and show that spatial structure is a stronger driver than global semantic accuracy.

Spatial vs global information tradeoff in pretrained vision encoders. Several recent works study the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). Siméoni et al. (2025) show that continued training of self-supervised vision representations can increase similarity between the global CLS token and patch tokens, which can hurt dense spatial tasks. Bolya et al. (2025); Heinrich et al. (2024) explicitly train spatially tuned models for dense spatial tasks. Our contribution brings this tradeoff into generative training: for representation alignment, the spatial structure of a vision encoder matters more than its global information, motivating future work on selecting and training external representations specifically for generation.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** We see that across different encoders - both spatial normalization layer and convolution projection layer (§4) lead to significant gains in convergence speed; with best results obtained using both. All results reported at 100K steps and SiT-XL/2.

Encoder	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **iREPA across training recipes.** Applying the spatial improvements on top of REPA-E (a) and MeanFlow with REPA (b) yields consistent gains, showing that the spatial-transfer contribution is not limited to the original REPA recipe. For MeanFlow, we report results at 1 and 4 NFes. CFG value of 2.0 is used for MeanFlow. All results are reported at 100K training iterations.

6 CONCLUSION

This paper revisits the mechanism behind representation alignment with a contribution structure centered on global information versus spatial structure. First, a large-scale empirical analysis shows that stronger global semantic accuracy does not reliably make a better REPA target. Second, spatial self-similarity metrics make the alternative measurable and correlate much more strongly with generation performance than linear probing. Third, iREPA turns the diagnosis into a minimal intervention: a convolutional projection and spatial normalization accentuate spatial transfer from target representations to diffusion features. Across encoders, model sizes, and training recipes, iREPA consistently improves REPA convergence. Together, these contributions argue that representation alignment should be treated as a spatial-transfer problem and that target representations for generation should be selected and designed accordingly.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **iREPA gains across encoders with and without classifier-free guidance.** These results show that the spatial-transfer contribution persists under CFG. SiT-XL/2 is used as the base model; Appendix E reports detailed SiT-B/2 and SiT-L/2 results across all encoders.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

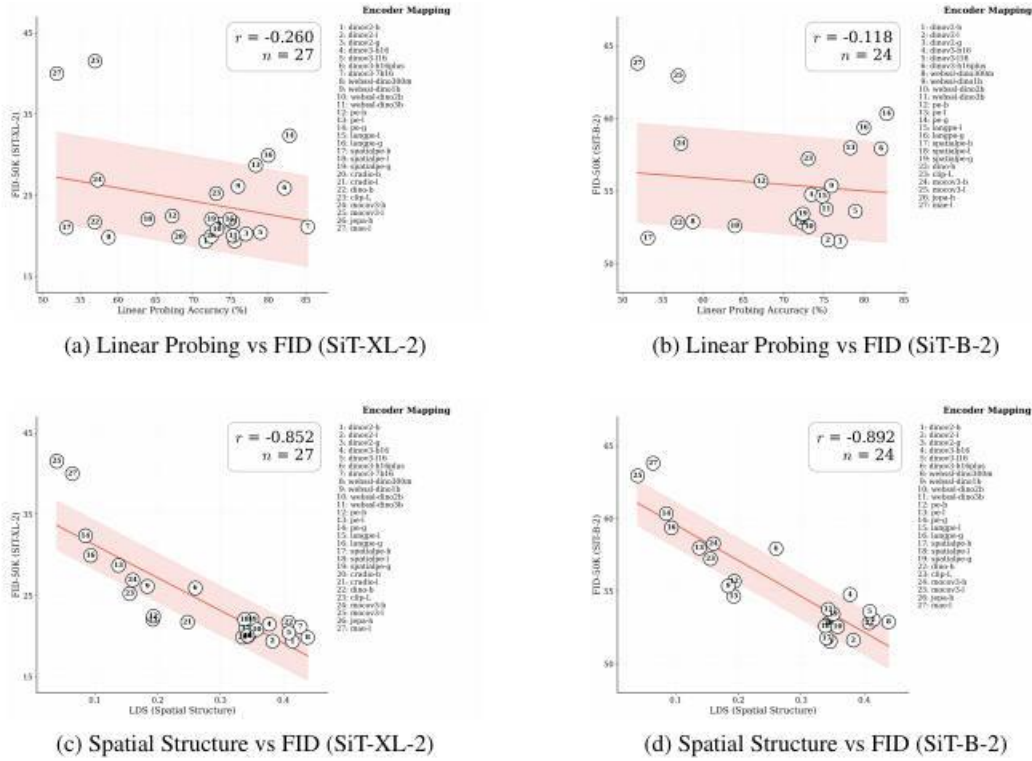


Figure 10: **Detailed correlation analysis with encoder labels.** Top row reports the baseline global-accuracy signal: linear probing accuracy vs FID for SiT-XL-2 and SiT-B-2. Bottom row reports the paper’s spatial contribution: LDS vs FID for the same model scales. Each point is labeled with its corresponding encoder number (see legend), making clear that spatial structure provides the stronger reviewer-facing explanation across scales.

B SPATIAL SELF-SIMILARITY METRICS

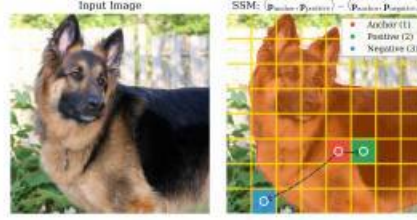


Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of SRSS metric: For each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within mask) and anchor-negative pairs (outside mask). Intuitively, larger values indicate that patches on the same semantic region are more similar than patches on unrelated regions — indicating better spatial structure preservation.

Setup. This appendix makes the SSM contribution fully reproducible. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the cosine self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We next specify the spatial metrics used to support the paper’s central claim that target-representation structure is more informative than global classification accuracy.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}].$$

where r_{near} and r_{far} are hyperparameters. By default, we use $r_{\text{near}} = r_{\text{far}} = H/2$, and the correlation is robust to their exact choices. Larger LDS values indicate the spatial organization central to our contribution: nearby patches are, on average, more similar to each other than distant patches.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate faster similarity decay with distance, providing another way to quantify the spatial organization that predicts REPA performance.

Semantic-Region Self-Similarity (SRSS). For each image, we first sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsampled to $H \times W$. We then select three types of points: anchor, positive, and negative. The anchor and positive points are sampled from within the mask (should have similar semantics), while the negative point is sampled from outside the mask (less related to anchor). Conceptually, if the encoder feature representation preserves the spatial structure, the anchor and positive points should have higher cosine similarity,

while the anchor and negative points should have lower cosine similarity. Thus, we define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

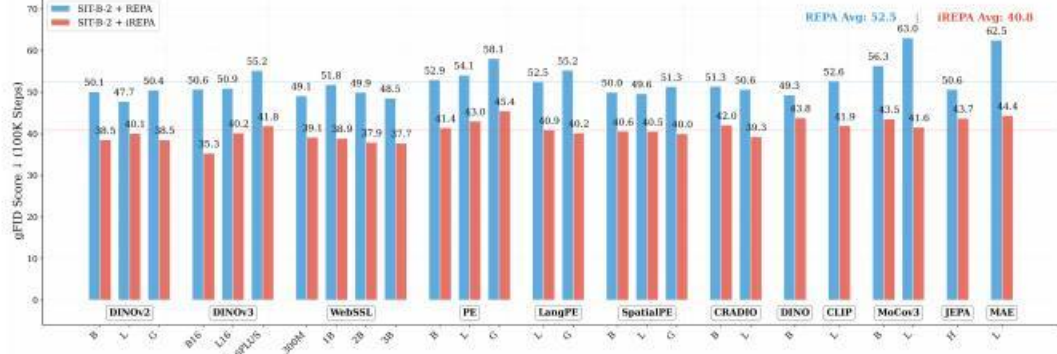
where positive points are within Manhattan distance $d \leq r_{\text{near}}$ from the anchor within the mask, and negative points are at distance $d \geq r_{\text{far}}$ outside the mask. This mask-based metric directly tests whether the representation preserves semantic-region structure.

RMS Spatial Contrast (RMSC). Finally, we consider a simple contrast metric to capture the spatial contrast between patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

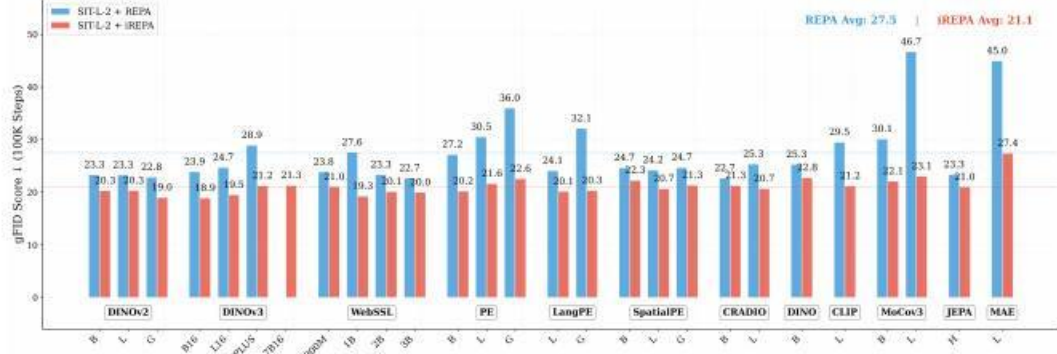
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representations and therefore stronger preserved spatial structure. Lower values indicate more uniform feature distributions, a failure mode consistent with global information diminishing spatial structure (Siméoni et al., 2025).

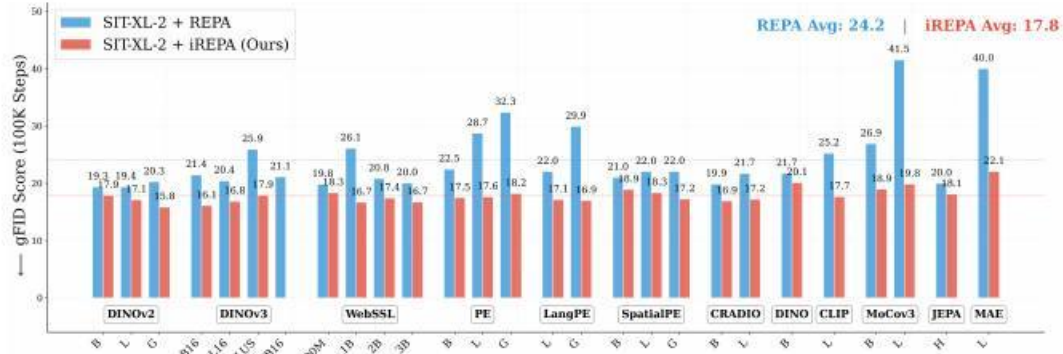
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



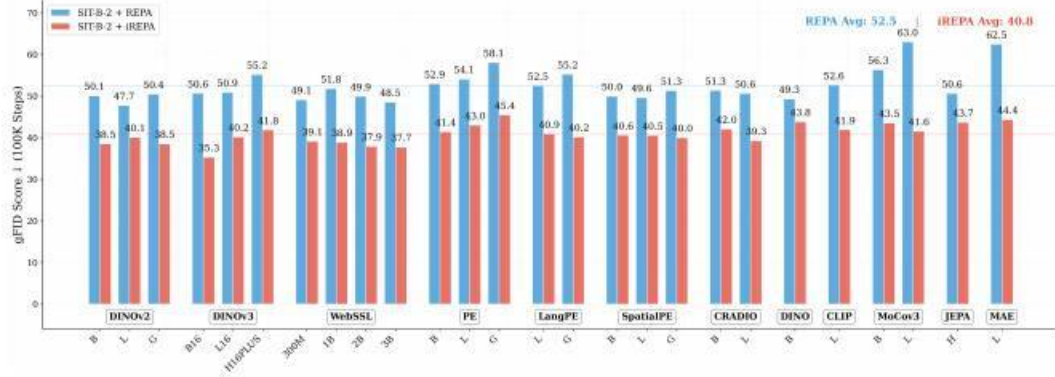
(b) SiT-L-2 (458M parameters)



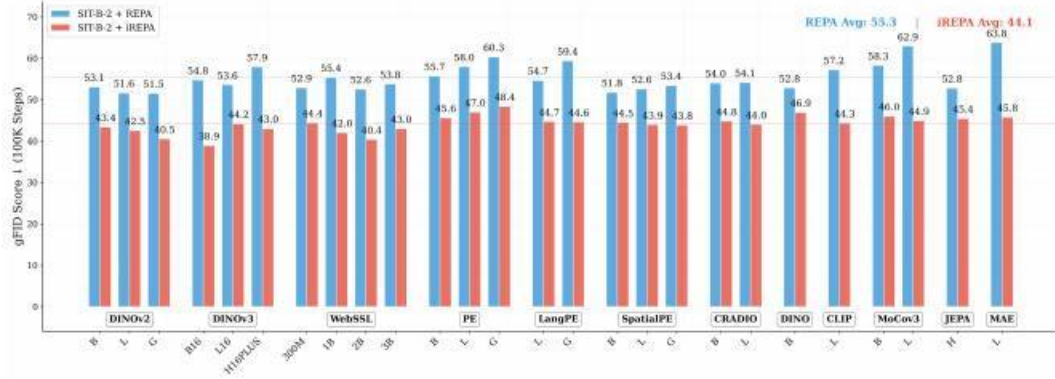
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies showing iREPA improvements across different vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** Comparison of SiT-B-2 performance with different encoder depths (6 vs 8 layers) across various vision encoders. The results show consistent improvements with iREPA regardless of encoder depth, with slightly better performance observed with 8 encoder layers. Both configurations demonstrate significant FID improvements when spatial regularization is applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across all encoders, with particularly strong gains in FID and IS metrics. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

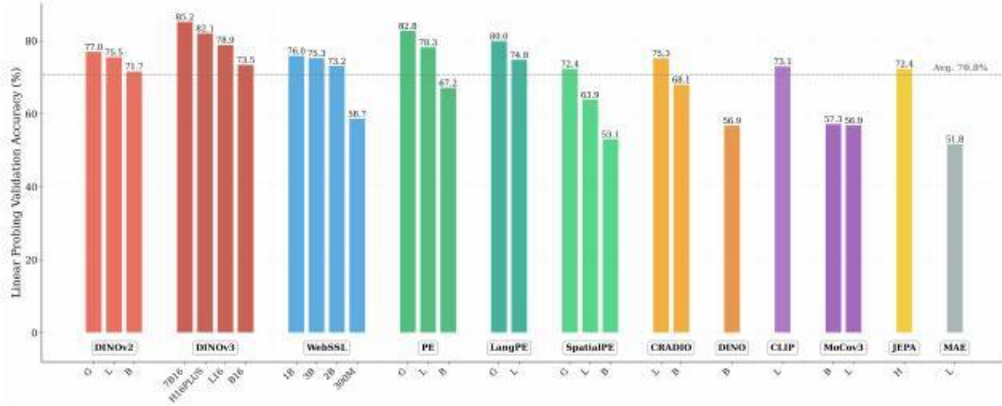


Figure 14: **Why global information is not the whole story.** In addition to the [CLS] token, the mean of patch tokens contains substantial global semantic information. This global component can improve global performance, but it reduces contrast between individual patch tokens and can hinder spatial-structure transfer. The iREPA contribution removes part of this global component (through the mean of patch tokens) to improve spatial transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA vs iREPA with CFG (scale 2.0) across different vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA consistently improves both IS and FID metrics across all encoders.

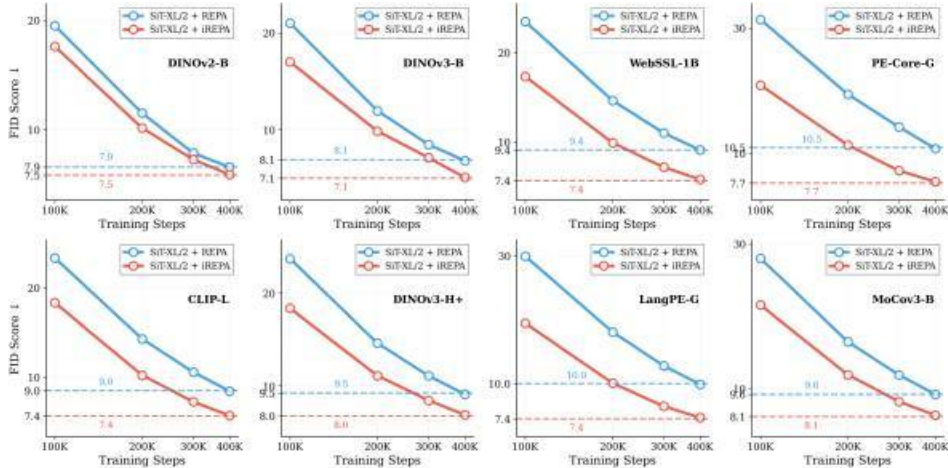


Figure 15: **Accentuating spatial features helps consistently improve convergence speed.** Results for SiT-XL/2 with REPA and iREPA.

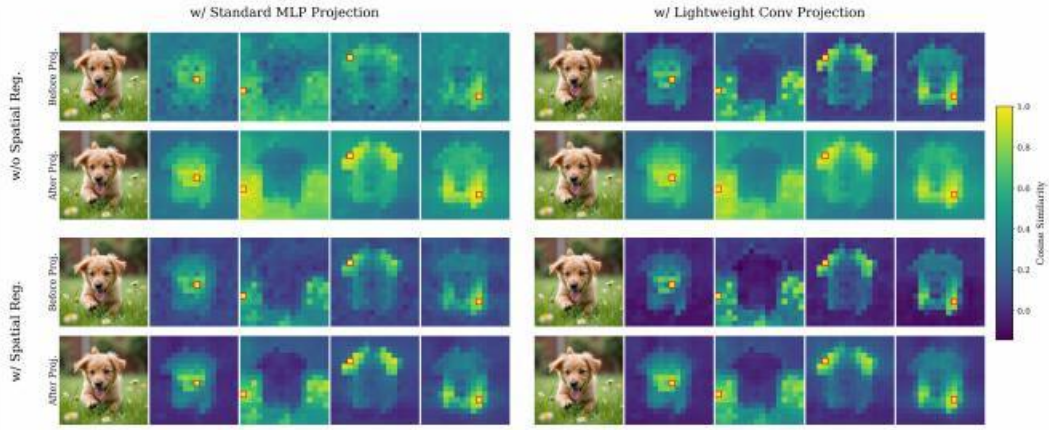


Figure 16: **Visualizing the iREPA contribution: two minimal changes enhance spatial feature transfer.** **First**, the standard MLP projection layer (top-left) loses spatial information while transferring features between the pretrained encoder (after projection) and diffusion model (before projection). A simpler convolution layer better preserves the spatial-transfer signal (top-right). **Second**, vision encoder features often have limited spatial contrast (§2), causing tokens in one semantic region (*e.g.*, dog) to have high cosine similarity with unrelated tokens (*e.g.*, background). A simple spatial regularization layer accentuates contrast in the learned representation and improves generation performance. The ablation confirms the contribution structure: best results are obtained when both components are used (Table 2).

G IMPLEMENTATION DETAILS

Training setup. We follow the REPA (Yu et al., 2024) setup unless otherwise specified so that the measured gains isolate the iREPA contribution. All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the ADM (Dhariwal & Nichol, 2021) protocol, center-cropping and resizing each image to 256×256 . We use the stabilityai/sd-vae-ft-mse VAE (AI, n.d.) throughout diffusion training and inference. For the spatial normalization layer, we use $\gamma \in [0.6, 0.8]$; for the projection layer, we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` for acceleration, along with gradient clipping and exponential moving average (EMA) for stable optimization.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a), we use the official open-source implementations. For MeanFlow (Geng et al., 2025), we use the implementation provided by the original authors, including REPA, and add only the two iREPA changes described in §4.

Evaluation. For image generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.), and recall (Rec.) (Kynkäänniemi et al., 2019), measured on 50K generated images. For sampling, we follow REPA (Yu et al., 2024) and use the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps following the official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Classical computer vision provides useful context for our spatial-structure contribution. Methods such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) were designed as robust local descriptors for object detection, matching, and related tasks. Recent generative-model work has instead emphasized modern self-supervised representations with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our results suggest a complementary selection principle: spatial coherence itself can make an alignment target useful, so both handcrafted descriptors and learned representations are relevant when they preserve strong spatial structure.

Pretrained visual encoders for generative models. Pretrained visual encoders support generative models in several ways: as discriminators that accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers for adversarial distillation in diffusion models (Sauer et al., 2024; Kang et al., 2024), and, more recently, as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features improve convergence and enable scaling to large datasets, as in StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. In diffusion models, adversarial distillation uses pretrained encoders to guide student networks toward higher-fidelity samples. REPA (Yu et al., 2024) shows that aligning diffusion features with external encoders improves convergence and generation quality. Our contribution refines this line of work by identifying spatial-structure preservation, rather than discriminative capability alone, as the key property to emphasize.

Denoising transformers. Transformer architectures are now central to scalable generative modeling, including diffusion transformers (Peebles & Xie, 2023) and flow-matching variants (Ma et al., 2024). Architectures such as GenTron (Chen et al., 2023) scale transformers beyond 3B parameters for text-to-image synthesis; U-ViT (Bao et al., 2023) shows that pure transformer backbones can be competitive without U-Net inductive biases; and ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) combine denoising with sequential generation to trade off quality and speed. These architectures remain expensive to train from scratch, motivating representation alignment methods such as REPA (Yu et al., 2024). Our contribution explains why such alignment accelerates training: the strongest signal is preserved spatial structure, which clarifies why some encoders accelerate training more than others and guides better alignment design. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very

significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denosing as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by [Abstreiter et al. \(2021\)](#) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA ([Hudson et al., 2024](#)) introduces a diffusion model with an information bottleneck to learn compact representations. [Chen et al. \(2024\)](#) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. [Wang & He \(2025\)](#) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, [Jiang et al. \(2025\)](#) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages learning of robust features. Our findings complement this view by showing that when diffusion models are aligned with strong spatial representations from self-supervised encoders, both the generative and discriminative capabilities improve, suggesting that spatial structure preservation is a key factor in this synergy.

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 ([Siméoni et al., 2025](#)) trains a 7B parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL ([Fan et al., 2025](#)) demonstrates that visual models trained on more than 2 billion images can match language-supervised models like CLIP ([Radford et al., 2021](#)) on vision-language tasks without language supervision. C-RADIO ([Heinrich et al., 2024](#)) combines multiple teacher models through distillation to create versatile encoders that excel across diverse visual domains. I-JEPA ([Assran et al., 2023](#)) explores predictive architectures that learn by predicting masked regions in representation space, instead of reconstructing pixels directly. SAM ([Ravi et al., 2024](#)) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family ([Bolya et al., 2025](#); [Cho et al., 2025](#)) shows that intermediate features often outperform final representations for dense prediction tasks. While these models are typically evaluated on global tasks like image classification, our work reveals a crucial insight: strong performance on discriminative benchmarks does not necessarily translate to better generation quality. Instead, we find that encoders preserving spatial structure, regardless of their classification accuracy—provide the most benefit for diffusion training. This suggests that the evaluation metrics for self-supervised encoders should be reconsidered when targeting generative applications, with spatial coherence being as important as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 ([OpenAI, 2025](#)) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code ([Anthropic, 2025](#)). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) use linear probing to measure the representation gap, using the increase in validation accuracy to explain the effectiveness of REPA. We find that spatial structure metrics (SSM) can also be used to measure the representation gap and explain the improvements across models. As shown in Figure 17, we see that representation alignment helps close the gap between SSM performance of a pretrained encoder like DINOv2 and the diffusion features.

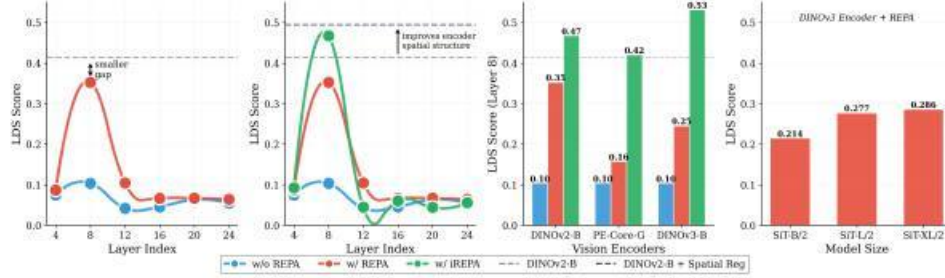


Figure 17: Representation alignment as a spatial-gap contribution. a) The contribution reframes representation alignment as reducing the spatial feature gap between diffusion and vision-encoder patch features. b) iREPA (§4) accentuates vision-encoder spatial features, at the cost of some global information, and improves generation performance. c) The method contribution improves performance across different vision encoders. d) The spatial-transfer gains scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

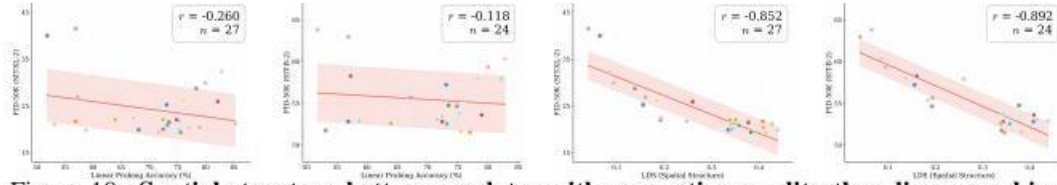


Figure 18: Spatial structure better correlates with generation quality than linear probing. Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) shows weak correlation. **Right two:** Spatial structure (LDS) vs FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) shows strong correlation. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

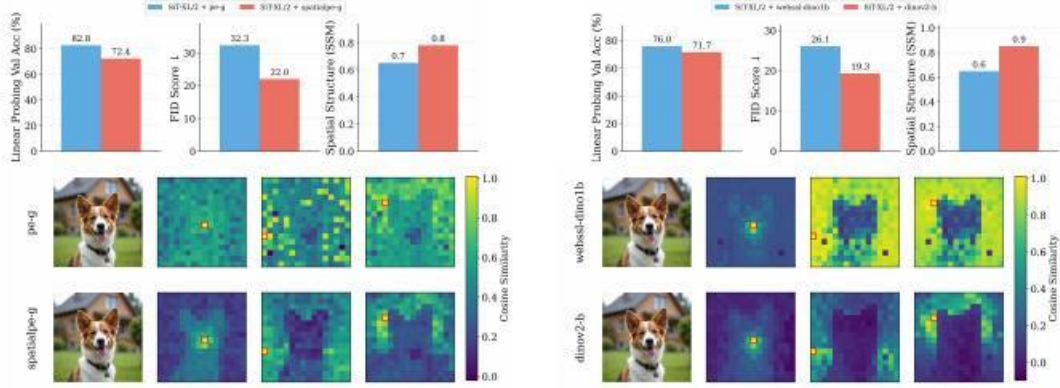


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b shows higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

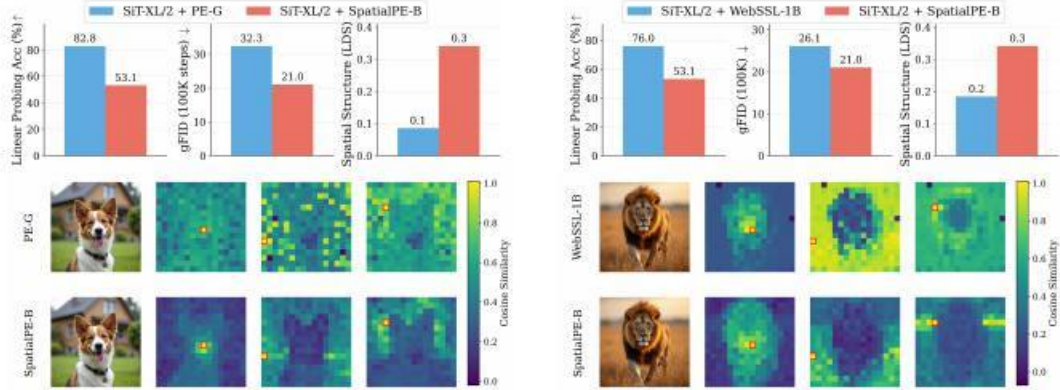


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the impact of spatial normalization on vision encoder representations. Vision encoders can have significant global components or overlays that limit the contrast between spatial tokens. For example, tokens in one semantic region can be highly correlated with tokens in unrelated regions (e.g., background), reducing the spatial distinctiveness of features. Spatial normalization removes this global overlay to enhance contrast between different spatial tokens, allowing the model to better preserve local spatial structure while reducing the dominance of global information that can interfere with generation quality.

The following figures demonstrate this effect across different examples, showing how spatial normalization transforms the feature representations to emphasize spatial structure:

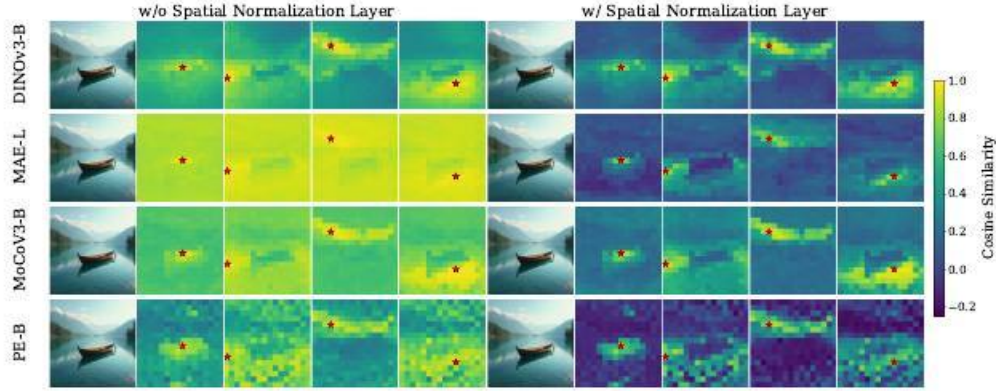


Figure 21: Visualizing impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

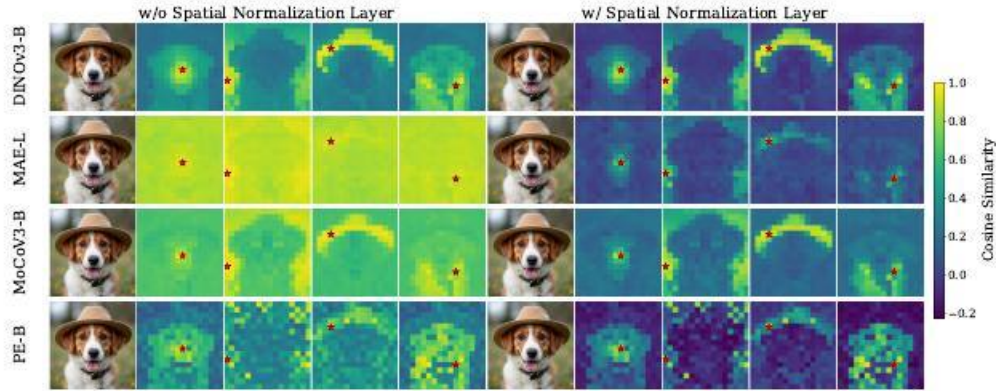


Figure 22: Visualizing impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

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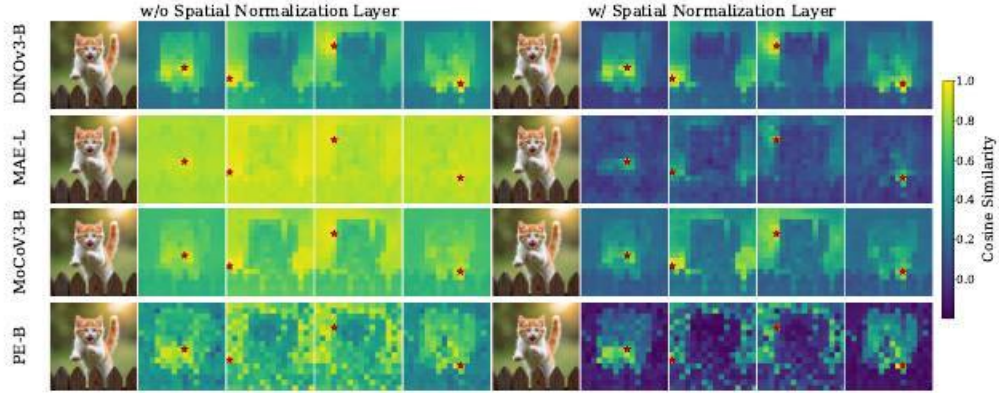


Figure 23: **Spatial-normalization mechanism (Example 2)**. Feature similarity maps show how spatial normalization supports the iREPA contribution by improving spatial contrast. Original features (left) contain a global overlay that blurs foreground/background distinctions; normalized features (right) are more locally coherent and better separated across semantic regions.

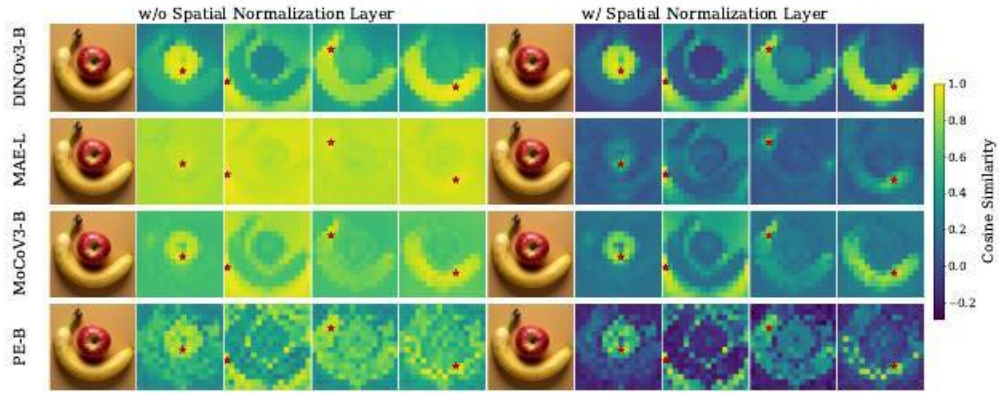


Figure 24: **Visualizing impact of spatial normalization (Example 4)**. The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct further experiments on ImageNet-512 (Deng et al., 2009) to evaluate the generalization of the proposed spatial improvements at higher resolutions. We follow the same setup as REPA (Yu et al., 2024), and report results across different choice of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.). All results are reported using SiT-XL and SiT-B using 50 NFE at inference w/o classifier free guidance. Fig. 25 shows the results with REPA before and after application of spatial improvements (iREPA). We observe that the spatial improvements also generalize to higher resolutions. Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.).

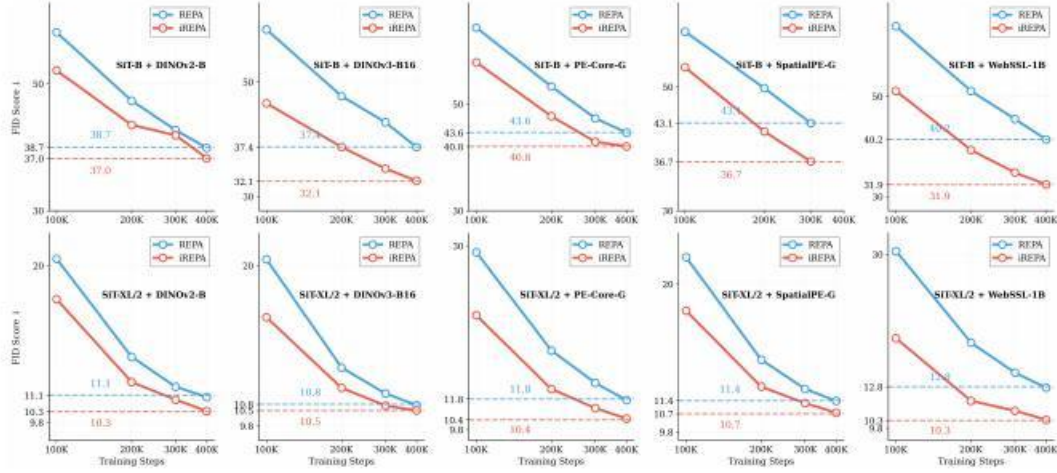


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features helps consistently improve convergence speed across different resolutions for both Imagenet 256 (Figure 7) and Imagenet 512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To further study the generalizability of the proposed spatial improvements beyond ImageNet, we also perform extensive experiments on multimodal T2I tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with various pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). Same as REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256, and evaluated using an ODE sampler with 50 NFE. Fig. 26 shows the results with REPA before and after application of spatial improvements (iREPA). We observe that the spatial improvements also generalize to multimodal T2I tasks. Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

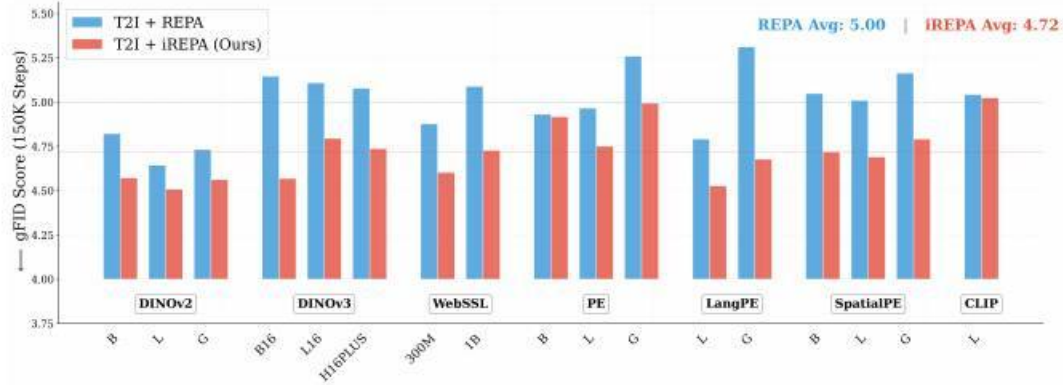


Figure 26: **Text-to-image generation across encoder variants.** Accentuating transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed for both imagenet (refer §4) and multimodal T2I tasks (above). Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 shows the updated correlations across different model sizes. We observe that spatial structure still shows much higher correlation with generation performance over linear probing accuracy. Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

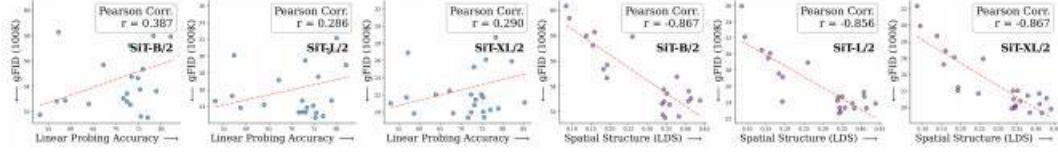


Figure 27: Correlation analysis without outliers. Across all model sizes (B, L, XL) spatial structure still shows much higher correlation (Pearson $|r| > 0.85$) with generation performance over linear probing accuracy (Pearson $|r| < 0.38$) after removing the outliers (MoCOv3-L and MAE-L). Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after application of spatial improvements with REPA. All results are reported using PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs) and with classifier-free guidance scale of 4.0. Similar to quantitative improvements (§4), we also observe that spatial improvements (iREPA) also help improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

To make the results easier to assess beyond standard metrics, we also test the robustness of the paper’s contributions with alternative generation-quality metrics (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). We use CMMD (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024), using the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021). These experiments ask whether both main claims—the SSM analysis from §3 predicts generation quality and the iREPA design from §4 improves spatial transfer—hold when FID is replaced by CMMD.

Spatial structure correlates much higher with generation performance than linear probing. We repeat the large-scale correlation analysis from §3 with CMMD (Jayasumana et al., 2024) across different vision encoders. Figure 29 reports SiT-B/2 results at 100K steps with REPA. Spatial metrics again correlate much more strongly with generation performance (Pearson $|r| > 0.88$) than linear probing (Pearson $|r| = 0.074$), showing that the SSM contribution from §3 is robust to the evaluation metric.

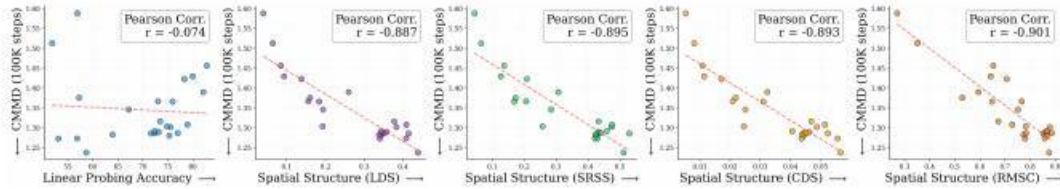


Figure 29: **Correlation analysis with CMMD metric.** Repeating the §3 analysis with CMMD (Jayasumana et al., 2024) instead of gFID preserves the same contribution-level conclusion. All results are reported using SiT-B/2 (100K steps) and REPA. Spatial metrics correlate much more strongly with generation performance (CMMD; Pearson $|r| > 0.88$) than linear probing (Pearson $|r| = 0.074$), showing that key empirical findings from §3 are robust to the evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We also evaluate the iREPA contribution from §4 with CMMD (Jayasumana et al., 2024) alongside standard metrics (gFID, sFID, IS, etc.). Table 8 shows that, across vision encoders, accentuating spatial transfer improves convergence under both CMMD and traditional metrics (IS, FID, sFID, Prec., Recall). The improvements hold both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We provide additional results with the CMMD metric (Jayasumana et al., 2024). All results are reported using SiT-XL/2, 256 batch size and traditional REPA as baseline. We adopt the pytorch implementation from (Paul, 2024) for computing the CMMD metric. Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. Across various vision encoders, accentuating transfer of spatial features (iREPA) helps improve convergence speed with both CMMD and traditional evaluation metrics (IS, FID etc.). This demonstrates that key empirical findings from §4 are robust to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 clarify how the spatial-normalization contribution from §4 behaves for SAM2-S (46M) encoder features². Spatial normalization improves SAM2 results, but the gains are marginal (Table 9). This is consistent with the method mechanism: spatial normalization removes a global patch-token component to enhance spatial contrast, while SAM2 already has little global information (validation accuracy < 24%), so there is less global component to remove (Figure 30).

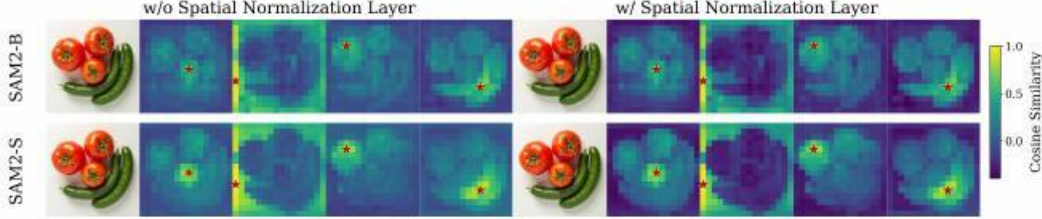


Figure 30: **Impact of spatial normalization on SAM2.** We observe that for SAM2, while use of spatial normalization layer does help enhance the spatial contrast, the improvements can be marginal. This is because spatial normalization (§4) relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **SAM2 spatial-improvement result.** iREPA tests whether the spatial-transfer contribution still helps an encoder with little global information. All results use SiT-XL/2 without classifier-free guidance, SAM2-S (46M) as the vision encoder, and traditional REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. As an additional robustness check for the SSM contribution, we repeat the correlation analysis from §3 with the validation accuracy after full-finetuning³ instead of linear probing. Results are shown in Fig. 31. All results are reported using SiT-XL/2 (100K steps) and REPA. For full-finetuning validation accuracy, instead of linear probing, we perform full-finetuning of vision encoder with learning rate of $5e-5$, warmup ratio of 0.1, and total of 30 epochs. Spatial metrics still correlate much more strongly with generation performance than full-finetuning accuracy, preserving the main diagnostic contribution under this alternative global-performance measure. Interestingly, gFID shows a weak positive correlation with validation accuracy after full-finetuning (Pearson $r = 0.317$): as validation accuracy increases, gFID increases and generation performance worsens. This reinforces the central contribution from §2: target representations with higher global semantic performance can show similar or worse generation performance under REPA.

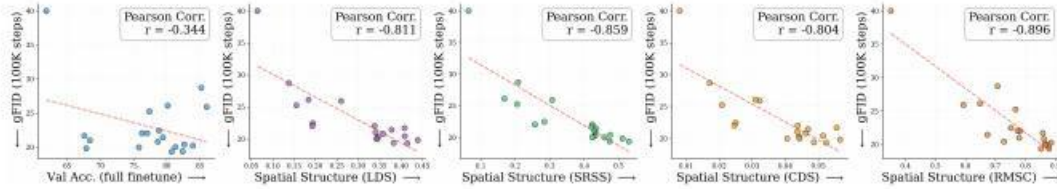


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. As an additional robustness check for the SSM contribution, we repeat the correlation analysis from §3 with the validation accuracy after full-finetuning instead of linear probing. All results are reported using SiT-XL/2 (100K steps) and REPA. Spatial metrics still correlate much more strongly with generation performance than full-finetuning accuracy, preserving the main diagnostic contribution under this alternative global-performance measure. Interestingly, gFID shows a weak positive correlation with validation accuracy after full-finetuning (Pearson $r = 0.317$): as validation accuracy increases, gFID increases and generation performance worsens. This reinforces the central contribution from §2: target representations with higher global semantic performance can show similar or worse generation performance under REPA.

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, when combined with JiT (Li & He, 2025a) iREPA outperforms recently proposed state-of-art pixel-space diffusion methods e.g., DeCo (Ma et al., 2025b). All results are reported with 200K training iterations with a batch size of 256 and evaluated using 50-step Euler sampling without classifier-free guidance. Results for DeCo are obtained directly from (Ma et al., 2025b).